

Chandan Kumar Singh

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Summary

Software Engineer with 1+ year experience designing scalable, secure payment-grade systems using Java, Spring Boot, Kafka, and AWS; delivered 40% latency reduction and 60% overhead cut through high-ownership engineering. Proven ability to build microservices, REST APIs, and event-driven platforms while applying SOLID principles and secure coding practices.

Experience

Tata Consultancy Services

Software Engineer — Java, Spring Boot, Microservices

Jan 2025 – Present

- Designed and owned RESTful APIs serving business-critical banking workflows; reduced average response latency by 40% and improved platform scalability by 50% through service-layer optimizations.
- Architected a centralized API Gateway with secret-key authentication and JWT token-reuse strategy, cutting inter-service overhead by 60% and improving distributed system reliability.
- Engineered secure integrations with external banking systems using AES-256 encryption and defensive patterns against MITM, Replay, and Request Tampering attacks, ensuring zero security incidents.
- Owned end-to-end delivery of internal automation solutions, eliminating 64% of manual operational effort and improving team release velocity.
- Engineered a Bulk Upload pipeline for concurrent ingestion of thousands of records per request, improving system throughput by 35% with zero data-consistency issues.

Projects

Clothify Shopping Application | Java, Spring Boot, Hibernate, PostgreSQL, Redis, Kafka, Docker, Kubernetes, AWS

- Designed and built a scalable e-commerce platform using microservices architecture (9+ services) with Java, Spring Boot, Hibernate, PostgreSQL, Redis, Kafka, Docker, Kubernetes, and AWS; applied SOLID principles and clean RESTful contracts for independent development and deployment.
- Implemented fault-tolerant inter-service communication via Feign Client with Circuit Breaker pattern, preventing cascading failures and improving system resilience.
- Integrated Redis for in-memory caching and Apache Kafka for asynchronous event-driven messaging, reducing service latency and decoupling dependencies.
- Achieved auto-scaling and high availability under variable traffic using AWS Lambda, API Gateway, Load Balancing, containerized with Docker and deployed on AWS.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Hibernate, Kafka, REST API, gRPC, FastAPI, React.js

Databases: MySQL, PostgreSQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, SVN, Jenkins, NumPy, Pandas, Scikit-learn, HuggingFace, Transformers, LangChain

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE)

Oct 2020 – Aug 2024